

AJT Question Bank

	Unit 1 : JDBC	Marks
1	Describe the following with proper illustrations: Type-4 JDBC driver.	5
2	Explain the JDBC URL with appropriate examples.	5
3	Describe the various categories of database drivers available. Also explain classes supported for establishing database connection with suitable examples.	10
4	Explain JDBC Architecture.	10
5	Explain types of various JDBC drivers. Using any one type of driver, explain how to create different statement handles. OR (All given alternatives same as original text)	10
6	Explain following classes: DriverManager, Connection, Statement, Resultset.	10
7	Write the steps to connect to the database in a java application.	10
8	What is a JDBC driver? What is its role? List the types of drivers and explain working of type-4 drivers. Give the different ways to create the JDBC connection with examples.	10
9	Which basic java package is used to interact with databases? List and explain some commonly used classes of this package.	10
10	Write a JDBC program to display contents of tbl_student on screen.	10
11	How do you make a selection of various JDBC drivers? Explain various types of drivers with examples.	10
12	Discuss CallableStatement with examples.	5
13	Explain the use of Callable Statement and Prepared Statement with example. OR (All alternatives same as original text)	10
14	What is a Callable statement? Write a procedure to insert a row into the table student (roll_no, name, age) and call it using a callable statement in JDBC application? OR (Alternative same as original text)	10
15	Explain the use of the Prepared Statement object of the JDBC with an appropriate example. OR Explain the use of PreparedStatement with appropriate examples.	10
16	Show the use of PreparedStatement objects to run precompiled SQL statements. Also write an example of a java snippet for PreparedStatement.	10
17	What is the use of Statement class in JDBC? List and explain various types of statements in JDBC.	10
18	Explain usage of following classes with respect to JDBC. Also explain some of their important methods. 1. Statement 2. Connection 3. Resultset	10
19	How does a Java application call Stored Procedures from DBMS? Explain with examples.	10
20	Explain use of DatabaseMetaData with examples.	5

21	Explain the use of JDBC and write the difference between JDBC and ODBC and write the program to create the database class that has one constructor which loads the JDBC driver and establishes the connection with the database and implement the methods i) <code>execSQLQuery(sqlstring)</code> ii) <code>execSQLUpdate(sqlstring)</code>	10
22	Explain how Transaction processing is handled by JDBC with example.	10
23	What is metadata? Explain metadata for resultset and database both. Explain the difference between <code>execute()</code> , <code>executeUpdate()</code> and <code>executeBatch()</code> method.	10
24	Write Java application program to change the <code>basic = basic + 500</code> of all the employees whose age is greater than 40 from employee table then display how many records updated.	10
25	Write a program using JDBC for getting personal information – name, birthdate, sex, address, phone no, email-id & store it in database. Also provide list of all records, all male, all female & all minors (age below 18).	10
26	Write database application that prints different records from a table. Find out the name of table, no. of columns, no. of records and type of the columns.	10
27	Consider Bank table with attributes AccountNo, CustomerName, Balance, Phone and Address. Develop a web application which allows insertion, updation and deletion of records in Bank table. Print values of all customers whose balance is greater than 20,000. Write servlet/JSP file and html file.	10
28	Write a JDBC program to insert and retrieve photo in database.	10
29	Write a database application to perform basic SQL operations using JDBC on table employee with attributes empno, name, designation, department and salary.	10
30	Write a difference between statement and prepared statement interface.	5
31	Write a sample code to store image in database.	5
32	Draw and explain J2EE architecture OR Explain J2EE Architecture and different Java EE components and containers. OR Draw and explain J2EE Architecture. Discuss the different Java EE components and containers.	10 IMP
33	Discuss all possible clients for J2EE application along with J2EE architecture.	10

Unit 2: Servlet

1	Differentiate the followings: any one three 1. GenericServlet vs HttpServlet 2. doGet() vs doPost() 3. Servlets vs CGI 4. Session and Cookie 5. JSP vs Servlet	5
2	Discuss the application design with MVC architecture.	5
3	Explain MVC architecture in detail. OR Write a short note on MVC architecture. OR Explain MVC architecture in detail with figures.	10
4	Discuss the use of GET and POST with example.	
5	Explain web Application, Web Container, Web Architecture. OR Explain the structure of a web application and the role of web containers.	10
6	Explain HTTP Protocol, Request and Response. Or Explain the concept of HTTP protocol and how a request is processed in a web application.	10
7	Give the characteristics of the HTTP protocol and explain the GET, HEAD and PUT methods of the HTTP protocol.	10
8	Explain Servlet Life Cycle in detail. OR Explain Servlet Life Cycle with example to demonstrate every state. OR Draw and explain Servlet life cycle. OR Explain the Servlet life cycle with life cycle methods.	5 OR 10
9	What is deployment descriptor? Explain various tags of DD. What is the use of DD?	10
10	List and explain various stages of Servlet life cycle. Explain role of web container.	10
11	Write sample deployment descriptor file (web.xml). Explain various configuration parameters of it.	10
12	What is a role of MVC in designing multi-tiered application? Explain technologies used at each tier.	10

13	Web Server Vs. Application Server	5
14	Write a servlet that redirect requests alternatively to JSP Pages named Oddand Even.	5
15	Explain use of ServletConfig and ServletContext object with example. OR Discuss the possible uses of the ServletContext and ServletConfig object.	5
16	What is Request Dispatcher? What is the difference between Requestdispatchers? forward () and include () method? OR Describe Request Dispatcher in Detail. Compare RequestDispatcher forward() and include() methods.	10
17	What is difference between Attribute and Parameter?	10
18	Explain ServletConfig, ServletContext, HttpSession. How it useful for any web application.	10
19	Explain Request and Response Objects in servlet.	10
20	Explain the concept of Servlet context.	10
21	Explain ServletContext life-cycle event handling with necessary code example.	10
22	Write a servlet RegistrationServlet to get the values from registration.html html page and display the contents. Write the web.xml file.	10
23	Write a Login servlet. Take input username and password from html file login.html and authenticate the user. Write the web.xml.	10
26	Discuss all possible clients for J2EE application along with J2EE architecture.	10
27	Write a Servlet program which maintains a counter for the number of times it hasbeen accessed since its loading.	10
28	Differentiate the followings: (Any Three) (IV) Session and Cookie	5
29	What is session? List the different ways to manage the session. OR Explain session tracking in Servlet and list different techniques used for session management.	5
30	What do you mean by Session? Write 4 session tracking method in detail.	10

31	Write JSP Page to demonstrate shopping cart using session feature of JSP.	10
32	What do you mean by Cookie? How to read and write with Cookie object.	10
33	What are cookies and sessions? How can you use them in Java EE? Compare them.	10
34	List the servlet session level events and show that how servlet destroy event is handled?	10
35	Explain various Servlet Events and its use. How to register listener for web app in Deployment Descriptor. (OR) Describe following with proper illustrations: Servlet Events. (OR) List out different types of servlet event and explain it.	10
36	What are cookies? Write a servlet that reads and prints all the previous cookies and add a cookie with your name.	10
37	Develop any Servlet application which demonstrates use of session management.	10
38	Explain event handling in Java Servlet with example(s).	10
39	What is Session? Define Session Tracking. What are the session tracking techniques?	10
40	How cookies are used to track a session? Explain with example.	10
41	Write a Servlet to set the cookies for first and last name entered by user.	10
42	Explain various Session tracking mechanisms in servlet with example.	10
43	What is Filter? List the applications of filter.	5
44	What do you mean by filter? What is the advantage of it? How to create a filter in your application?	10
45	What are servlet filters? Give the necessary API for filters and explain their use.	10
46	What is filter? What is its use? List the different filter interfaces with their important methods.	10
47	Enlist and explain the need of filters with a program.	10
48	What is doFilter() method? What are its parameters? Give its use with proper example.	10
49	What is the use of filter in Servlet? Explain with suitable	10

	example.	
50	Write a servlet which counts the number of digits into an integer received as parameter. Give the necessary web.xml file to deploy the servlet.	10
51	Write a servlet which accept two numbers using POST methods and display the maximum of them.	10
51	Write a web application using servlet to compute an area of a circle. Get the radius from the client. Write necessary web.xml file.	10
53	What is servlet configuration? Show it with example web.xml file.	10
54	Write a Web application using servlet to find the sum of all the digits of an input integer.	10
55	Write a Java Servlet to print BE Semester 7 Marksheet of entered enrollment number by user using JDBC Write all the files.	10
56	Write servlet which displayed following information of client. I. Client Browser II. Client IP address III. Client Port No IV. Server Port No V. Local Port No VI. Method used by client for form submission VII. Query String name and values	10
57	Write Servlet program to display "Hello world" on screen.	10
58	Design a form to input details of an employee and submit the data to a servlet. Write code for servlet that will save the entered details as a new record in database table Employee with fields (EmpId, Ename, Email, Age) write html, servlet and web.xml files.	10
59	Write Servlet program to create cookie. Also write code to display contents of cookie on page.	10
60	How are the requests serviced by Servlets? Discuss Servlet life cycle methods.	10
	Unit 2: Java Server Pages	
61	Differentiate the followings: (Any Three) (V) JSP vs Servlet OR Compare JSP and Servlet.	5

62	List JSP implicit objects and explain any two.	3
63	Explain JSP Life Cycle in detail. OR Briefly explain JSP life cycle. OR List and explain various stages of JSP life cycle. Briefly give the function of each phase. OR Draw and explain JSP page life cycle. OR Explain JSP architecture and JSP page life cycle.	5 OR 10
66	OR List & explain various stages of JSP life cycle.	
67	Draw and explain JSP architecture.	10
68	Compare JSP with Servlet.	10
69	Write all Implicit Objects of JSP in detail. Or Explain the JSP implicit objects with their use. OR Show the use of JSP inbuilt objects: request and response, with their use in application. OR Enlist and explain usage of various implicit JSP objects. OR List & explain implicit objects in JSP. OR Enlist JSP implicit objects. Explain any three of them. OR Explain JSP elements and JSP implicit objects with examples.	10
70	Explain JSP Object scope: (i) Page (ii) Request (iii) Session (iv) Application with example.	10
71	Write a JSP page that prints your resume in well formatted form	10
72	Write a JSP page to display your semester mark sheet. Give the necessary files to deploy it.	10
73	Develop a JSP page to display student information with subjects for a particular semester from the database.	10
74	Explain the given attributes of the JSP directive with an example. 1. isErrorPage 2. Import 3. contentType 4. language	5

75	Explain use of <jsp:useBean> action tag with example.	5
76	Explain 1) Scriptlet 2) Declaration 3) Expression 4) EL Expression in detail.	10
77	Explain directive tag. And types of directives in JSP in detail.	10
78	Show the use of JSP page directive tag with its attributes. OR Explain JSP (i) page (ii) include (iii) taglib directive with attribute, description. Also write example of java snippet for each directive.	10
79	Explain all JSP Action in detail.	10
80	Explain the following with respect to JSP Unified EL. i. Value expressions ii. Method expressions	10
81	Discuss the Expression Language feature of JSP and mention various implicit objects available in the EL with description.	10
82	Give an example showing the use of EL functions.	10
83	Explain the action tags used to access the JavaBeans from a JSP page with example.	10
84	Enlist and explain the purpose and use of action tags in JSP.	10
85	Write small web application which takes marks of three subject and pass to servlet. Servlet forward to model class having method getClass() and getPercentage(). Display class and percentage in .jsp page.	10
86	Write a student bean class with property student_name, enrollment_no, address and cpi. Write jsp page to set and display all property.	10
89	Explain usage of JSP Page directive with various attributes of it.	10
90	Write a program to create HTML form with one text box to enter number in it. When page is submitted to JSP, it should print entered number is odd or even.	10
91	Explain the three ways of including a JSP in another JSP with example.	10
92	Write a java bean named "student" having roll no and name having getter & setter methods. Write a JSP page to set the roll number and name for a student object and then print them by reading from object.	10

93	Write a web application using JSP to get an integer N(N>0) and display all the prime numbers up to N.	10
94	Explain the various types of EL expression with example.	10
95	Differentiate Java Beans and Simple Java Class	5
96	Explain JSP directives with examples	10
97	What is session object in JSP? Explain it with example.	10
98	Write a JSP program using JSTL SQL taglib to display student details in tabular form by iterating through the database table student.	5
99	List the JSTL core tags and explain any four with example.	5
100	Explain various tags in JSTL with example: (i) Core (ii)XML	10
101	Explain various tags in JSTL with example: (i)SQL (ii) Functions	10
102	Explain JSTL in detail.	10
103	Explain the features of the core tag library for JSP. OR Explain JSP tag library with examples.	10
104	Write a note on SQL tag library.	10
105	Write a note on the CORE tag library. OR Explain JSTL core tags.	10
106	What is the XML tag library? Explain the XML core tags and show their use.	10
107	Explain transaction handling using JSTL.	10
108	Write a JSTL program to parse & display text of XML file.	10
109	Write a login.jsp page to get the login information from the user and authenticate the user with the database with JSTL SQL tags.	10
110	Write and explain various tags of JSTL core and sql library.	10
111	Write line(s) of code in JSP for the following. i. Session read and write ii. URL rewriting sending and retrieving parameter(s) iii. URL redirection iv. Print "hello world" as output v. Include the other JSP file statically vi. Expression to display date as output vii. Method of setting the JSP parameters to use in JSTL	10

112	Write a program to create login form using HTML. When a form is submitted to JSP, fetch submitted details and check whether the user is valid or not? (Userdetails can be hardcoded. No need for a database).	10
113	Explain JSP Core Tag library and SQL Tag library with examples.	10

	UNIT 3: Java Network Programming	
	Explain the basic concept of network programming and the role of Socket in Java.	
1	Explain the following classes with their use. i. URLConnection class	5
2	Use of the URL class.	5
3	Explain the function of the following classes: (1) URL (2) URLConnection OR Outline and analyze the functional differences between URL and URLConnection classes.	10 IMP
4	Write a short note on the Protocol handler.	10
5	Explain the following classes with their use. ii. DatagramSocket and DatagramPacket class	5
6	Give differences between Socket class versus ServerSocket class	5
7	Write a client program to send any string from its standard input to the server program. The server program reads the string, finds the number of characters and digits and sends it back to the client program. Use connection-oriented or connection-less communication.	10
8	Write a client-server program using TCP sockets to echo the message sent by the client.	10
9	What is Server Socket? How does it work in java? Explain it with an example. OR What is Server Socket? Explain in detail with an example.	10
10	What is Datagram Socket? Explain in detail with examples.	10
11	Write a TCP or UDP client and server program to do the following: client> java client localhost/IP Port <enter> Enter text: This is my text to be changed by the SERVER <enter> Response from server: reverses EHT YB DEGNAHC EB	10

	OT TXET YM SI SIHt client> exit	
12	Write an UDP client and server program to do the following: Client send any string and server respond with its capital string.	10
13	Write a client server program using TCP where client sends two numbers and server responds with sum of them.	10
14	Write a Java Client and Server Program to get date time from server on the client request. OR	10
15	Write a TCP or UDP client and server program to do the following: client> java client localhost/IP Port Enter text: Welcome to Gujarat Technological UNIVERSITY Response from server: ytisrevinu LACIGOLONHCeT TARAJUg TO EMOCLEw client> exit	10
16	Write a client-server program using TCP or UDP where the client sends 10 numbers and server responds with the numbers in sorted order.	10
17	Write a UDP Client-Server program in which the Client sends any string and Server responds with Reverse string.	10
18	Write a client server program using TCP where client sends a string and server checks whether that string is palindrome or not and responds with appropriate message.	10
19	Write a client-server program using UDP socket. Client send list of N numbers to server and server respond the sum of N numbers.	10
20	Write a client-server program using UDP socket. Client sends the list of N strings and server responds the concatenation of those strings.	10
21	Write java network program using network API in which client sends one string to server and server replies with length of string.	10
22	Write a connection-oriented client server application where the client sends any string for its standard input and the server program responds back with the reversed string.	10
25	Write a program using JDBC for getting personal information – name, birthdate,sex, address, phone no, email-id & store it in database. Also provide list of all records, all male, all female & all minors (age below 18).	10
26	Write database application that prints different records from a table. Find out the name of	10

	table, no. of columns, no. of records and type of the columns.	
27	Consider Bank table with attributes AccountNo, CustomerName, Balance, Phone and Address. Developed a web application which allows insertion, updation and deletion of records in Bank table. Print values of all customers whose balance is greater than 20,000. Write servlet/JSP file and html file.	10
28	Write a JDBC program to insert and retrieve photo in database.	10
29	Write a database application to perform basic SQL operations using JDBC on table employee with attributes empno, name, designation, department and salary	10
30	Write a difference between statement and prepared statement interface	5
31	Write a sample code to store image in database	
32	Explain TCP/IP client and server sockets using Java.	
33	Explain Java Mail architecture and protocols used in Java Mail.	
34	Define ServletConfig and ServletContext objects and list their uses with examples.	10 IMP
35	Explain the use of the Prepared Statement object of the JDBC with an appropriate example.	10 IMP
	UNIT 3 : Hibernate	
1	What is hibernate? List the advantages of hibernate over JDBC.	5
2	What is Hibernate? What are the advantages of it? Explain the architecture of it in detail. OR Explain the Hibernate architecture and Object Relational Mapping (ORM). OR Explain architecture of Hibernate. OR Draw and explain the architecture of Hibernate. OR Explain Hibernate Architecture and Object Relation Mapping in Hibernate with java code and required XML files.	10
4	Explain the Hibernate cache architecture.	10
5	Write a note on Hibernate technology.	10
6	Develop a program to get all students' data from a database using hibernate. Write necessary xml files.	10
7	What are various java technologies to connect with Databases? Explain Hibernate architecture for object-relational mapping.	10

8	What is O/R mapping? What is HQL? What are the advantages of HQL? OR Analyze the concept of O/R mapping and HQL in Hibernate ORM.	10 IMP
9	Explain the OR mapping in hibernate with an example. OR What is O/R Mapping? How it is implemented using Hibernate. Explain with examples. OR Explain O/R Mapping in hibernate.	10
10	What is HQL? How does it differ from SQL? Give its advantages.	10
11	What is OR mapping? Give an example of a Hibernate XML mapping file.	10
12	Explain Hibernate annotations and Hibernate Query Language (HQL).	
Unit 4: Spring MVC		
1	What is Dependency Injection?	5
2	What is Spring IoC container?	5
3	Briefly explain spring bean life cycle.	5
4	What is Spring? What are the advantages of Spring framework? What are features of Spring?	10
5	How many modules are there in Spring? What are they?	10
6	What are the types of Dependency Injection Spring supports?	10
7	What is Bean Factory? What is Application Context? What is the difference between Bean Factory and Application Context?	10
8	What are the common implementations of the Application Context? What does a typical spring implementation look like?	10
9	What is the typical Bean life cycle in Spring Bean Factory Container?	10
10	What is Spring's JdbcTemplate? What is PreparedStatementCreator?	10
11	What is IOC (or Dependency Injection)? What are the different types of IOC (dependency injection)? What are the benefits of IOC (Dependency Injection)?	10
12	What are Bean scopes in Spring Framework? Write use of each.	10
13	What is Spring Web MVC framework? List its key features.	10
14	Explain architecture of Spring MVC Framework. Explain all modules in brief.	10

15	What is AOP? What are the advantages of spring AOP? What are the AOP terminology?	10
16	What are the types of the transaction management Spring supports? What are the	10
	benefits of the Spring Framework transaction management?	
17	Explain about the Spring DAO support? What are the exceptions thrown by theSpring DAO classes?	10
18	Develop a small application using Spring MVC framework.	10
	Unit 4: Struts	
1	What is the difference between Struts and Struts2?	5
2	Describe various components of the Struts framework.	10
3	Draw and explain Basic MVC Architecture	10
4	Draw and explain Struts 2 - Architecture	10